Experiment No: 09

```
#include <stdio.h>
#include <stdlib.h>
#include<conio.h>
#include<time.h>
#include<math.h>
#define   size_x  320
#define   size_y  384
#define   smooth  3
typedef struct{
            int imagesize_x, imagesize_y;    int **pixel; }  image_t;
      image_t allocate_image(const int imagesize_x, const int imagesize_y);  int mod(int z, int l);
            int mod(int z, int l);

int main()  {
  image_t   image_in,image_out;
  int m,n, temp;    int smooth,csx,csy;    int k,l,x,y,sum1,sum2;   FILE *file_in,*file_out;
  char dummy[50]="";
  file_in=fopen("i9.pgm", "r+");
  file_out=fopen("avg_smooth_image.pgm", "w");
  fgets(dummy,50,file_in);
  do  {  fgets(dummy,50,file_in); }  while(dummy[0]=='#');  fgets(dummy,50,file_in);
  fprintf(file_out,"P2\n%d %d\n255\n",(size_x),(size_y));

  image_noise = allocate_image(size_x,size_y);
  image_filtered = allocate_image(size_x,size_y);
```

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```
   for (n = 0; n < size_y; n++)  {
     for (m = 0; m <size_x; m++)     {
                        fscanf(file_in,"%d",&image_noise.pixel[m][n]);    }  }

  csx=smooth/2;
  csy=smooth/2;
   for (n = 0; n < size_y; n++)  {
    for (m = 0; m < size_x; m++)  {
      sum1=0;
      for(k=0;k<smooth;k++)   {
        for(l=0;l<smooth;l++)        {
                  x=m+k-csx;        y=n+l-csy;
                 sum1+=image_noise.pixel[mod(x,size_x)][mod(y,size_y)];
        }    }
      image_filtered.pixel[m][n]=sum1/(smooth*smooth);
     }   }

   /*   Writing image (pgm) file  */
   for (n = 0; n < size_y; n++)  {
     for (m = 0; m <size_x; m++)     {
                        fprintf(file_out,"%d ",image_filtered.pixel[m][n]);
     }  }  fclose(file_out); }

 /* periodic extension (outside of the  image frame) */
 int mod(int z, int l)  {
      if( z >= 0 && z < l )     return z;
      else if( z < 0)           return (z+l);
      else if( z > (l-1))       return (z-l);  }

 image_t allocate_image(const int imagesize_x, const int imagesize_y) {
   image_t result;
   int x =  0, y = 0;

   result.imagesize_x = imagesize_x;
   result.imagesize_y = imagesize_y;

   result.pixel =(int **) calloc(imagesize_x, sizeof(int*));

   for(x = 0; x < imagesize_x; x++)  {
     result.pixel[x] =(int*) calloc(imagesize_y, sizeof(int));
     for(y = 0; y < imagesize_y; y++)
        {   result.pixel[x][y] = 0;    }
   }
   return result;
 }
```

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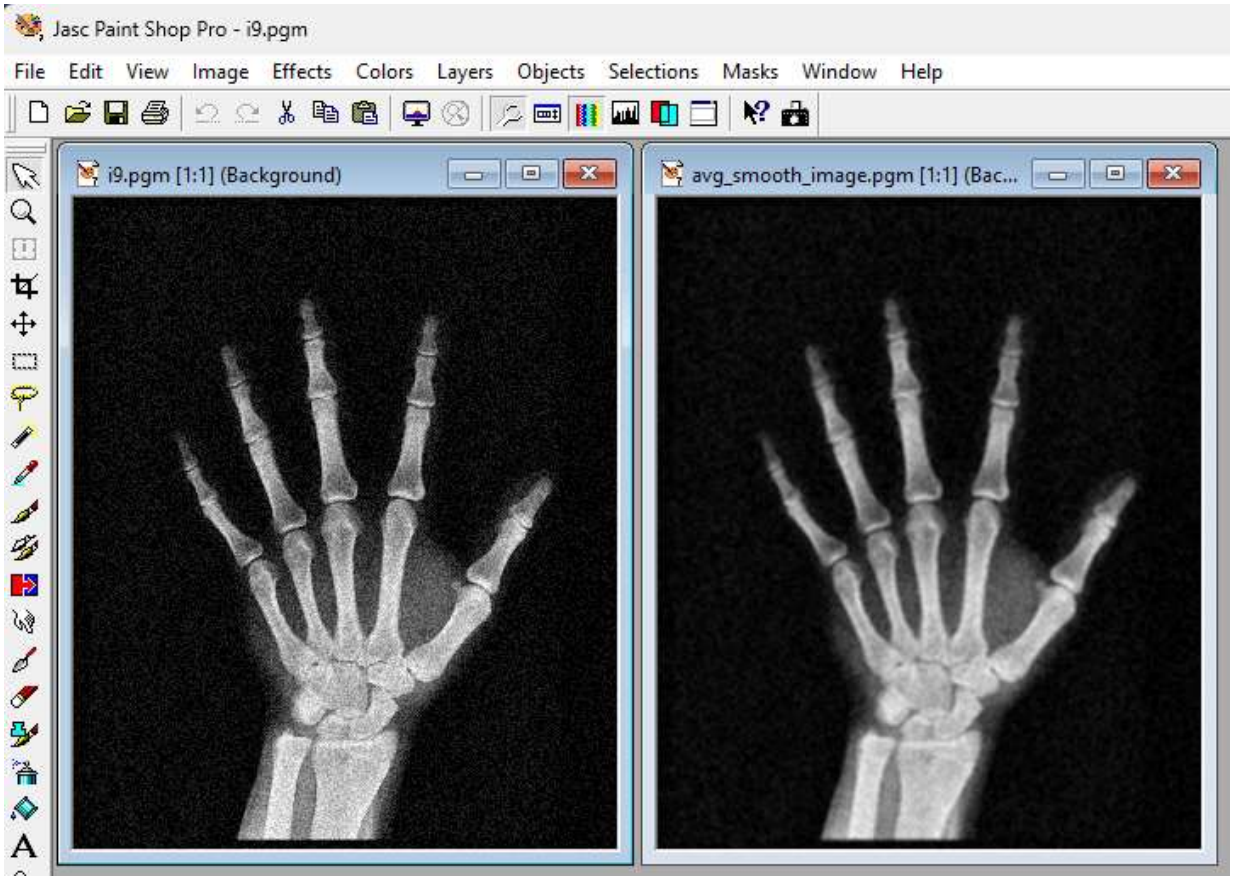

Left-side-Figure 9(a): Noisy image (i9.pgm)
Right-side-Figure 9(b): Noisy removed image (avg_smooth_image.pgm)